

BirdCLEF 2026

Modeling Hypotheses & Strategy Report

Synthesis of EDA Sections A–I
Phase 6 — Ranked Modeling Ideas & Hypotheses
Generated: 2026-04-12

Stat	Value
Target species	234 (162 birds, 35 frogs, 28 insects, 8 mammals, 1 reptile)
Training clips	35,549 (206 of 234 species have clips)
Zero-clip species	28 — only supervised via 66 labeled soundscape windows
Train soundscapes	10,658 total — only 0.6% have expert window labels
Prediction unit	5-second windows → 234-class probability vector
Evaluation metric	Macro-averaged ROC-AUC (skip classes absent in test)
Inference constraint	CPU-only Kaggle notebook, 90-minute hard limit
Domain gap	Global XC/iNat clips → Pantanal passive recorders (severe)
Competition deadline	June 3, 2026 (~8 weeks remaining)

This report synthesizes all EDA findings into a ranked, evidence-grounded set of modeling hypotheses, data pipeline recommendations, and a phased experiment plan. Every recommendation traces back to a specific EDA finding.

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§1 — Competition Structure: What the Model Must Actually Do

Before any modeling hypothesis can be tested, the prediction contract must be understood precisely. Misunderstanding the supervision unit vs. the scoring unit is the most common source of wasted effort in BirdCLEF-style competitions.

1.1 Prediction Contract

Axis	Training side	Scoring side
Unit	Whole clip (3–120 s)	5-second window
Label type	Clip-level primary + secondary	Per-window presence/absence
Label count	Mostly 1; 12.3% have secondaries	3–5 species per window (EDA Section F)
Supervision	35K clips + 1,478 soundscape windows	Hidden test soundscapes
Geography	Global (lat –55 to +70)	Pantanal, Brazil only
Recorder	Close-mic / phone / field recorder	Passive ARU deployment

Core tension: The model is trained on close-mic clips with whole-clip labels and must generalize to passive soundscape recorders predicting at 5-second resolution. This is a three-way mismatch: recording device, label granularity, and geographic domain. Every design decision must close at least one of these gaps.

1.2 Metric Implications

- **Macro ROC-AUC** — each of 234 classes gets equal weight regardless of frequency. A species appearing in 2 test windows matters as much as one appearing in 2,000.
- Metric is **threshold-free** — calibration of absolute probability values does not matter. Relative ranking within each class is what counts.
- **Classes absent in test are skipped** — but you cannot know which ones. Do not deprioritize rare species.
- Improving common-species AUC by sacrificing rare-species AUC is net-neutral at best and often net-negative under macro averaging.
- This metric strongly rewards models that *discriminate* within rare classes, not just models that avoid false positives overall.

1.3 CPU Inference Constraint (90-minute, no GPU)

- Hard limit on model size. EfficientNet-B4 and below are feasible; B6+ and large ViTs will time out.
- Mel spectrogram extraction must be batched — per-window librosa calls add ~2× overhead.
- ONNX export or torch.compile will likely be necessary for competitive inference speed.
- This constraint is known from BirdCLEF 2024 and 2025. Top solutions used EfficientNet-B0 to B2 or small ConvNext variants with ONNX export.
- Plan model architecture around inference budget first, then optimize within that envelope.

§2 — Ranked Modeling Hypotheses (Top 12)

Each hypothesis is grounded in a specific EDA finding. Priority is assigned by **expected AUC impact** × **implementation feasibility**. Hypotheses are grouped into tiers: Critical (must test first), High (test in first 2 weeks), and Medium (test after baseline is established).

Tier 1 — Critical (Expected AUC impact: High)

#	Hypothesis	EDA Evidence	Expected Impact
H1	Background mixing with unlabeled Pantanal soundscapes	Section 5.1 of EDA report	High
H2	Waveform mixup ($\lambda \sim \text{Beta}(0.4, 0.4)$) with soft multilabel loss	Section 4.2 of EDA report	High
H3	Clip windowing with energy gating reduces label noise	Section 5.3 of EDA report	High
H4	A validation scheme that separates by recording site	Section 3.2 of EDA report	High

Tier 2 — High Priority (Expected AUC impact: Medium–High)

#	Hypothesis	EDA Evidence	Expected Impact
H5	BirdCLEF 2025 Pantanal clips and soundscapes	Section 5.2 of EDA report	Medium
H6	Backbone pre-training on all available BirdCLEF clips	Section 4.3 of EDA report	Medium
H7	Per-clip peak normalization before mel extraction	Section 5.4 of EDA report	Medium
H8	A multi-taxon head architecture (shared backbone)	Section 4.4 of EDA report	Medium

Tier 3 — Medium Priority (Test after baseline is stable)

#	Hypothesis	EDA Evidence	Expected Impact
H9	Semi-supervised learning on the 10,592 unlabeled recordings	Section B.4.1.592 pseudo-labeling on unlabeled recordings	Medium impact, provides additional data for training
H10	Time-of-day conditioning (sin/cos encoding of hour)	Section B.4.1.592 pseudo-labeling on unlabeled recordings	Medium impact, provides additional data for training
H11	The 25 insect sonotypes (parent taxon 47158) represent 25 insect sonotypes	Section B.4.1.592 pseudo-labeling on unlabeled recordings	Medium impact, provides additional data for training
H12	Post-processing the output distribution by species	Section B.4.1.592 pseudo-labeling on unlabeled recordings	Medium impact, provides additional data for training

Priority summary: H1 (background mixing) → H2 (waveform mixup) → H3 (energy gating) → H4 (site-stratified CV) → H7 (normalization) → H5 (BC2025 data) → H6 (backbone pre-train) → H8 (multi-taxon head) → H9 (semi-supervised) → H10 (time-of-day) → H11 (sonotype 2-stage) → H12 (post-processing)

Do not test H9–H12 until H1–H4 are verified on OOF CV. Stacking unvalidated interventions produces uninterpretable results.

§3 — Data Pipeline Architecture

3.1 Input Sources (Ranked by In-Domain Value)

Source	Size	In-Domain?	Labels?	Recommended Use
Train soundscapes (BC2026)	10,658 ogg	YES — Pantanal	66 labeled (1,478 species)	Primary training + all as backgrounds
Train clips (BC2026)	35,549 ogg	Partial — global	Clips level	Training with windowing + energy gating
BC2025 clips + soundscapes	~50K estimated	High — same region	Clip + soundscape	Additional training after dedup
Prior BC clips (2021–2024)	Large	Low — different region	Clip level	Backbone pre-training only
Additional Neotropical XC	Variable	Medium — regional	Clip level	For rare/zero-clip species only
Non-Pantanal soundscapes	Various	NO — different domain	Various	Do NOT use as domain training data

3.2 Clip Processing Pipeline

Step	Action
[1] Load & resample	Load OGG → resample to 32 kHz mono. All clips already 32 kHz, minimal resampling needed.
[2] Peak normalization	Normalize peak to 0.9. Removes 600× loudness variance. Apply before any windowing.
[3] Window extraction	Slice into non-overlapping 5s windows. Short clips (<5s): pad with zeros OR treat as single window.
[4] Energy gating	Compute RMS of each window. If RMS < −45 dBFS → mark as background_only (no species label). Do NOT assign species labels to background windows.
[5] Background mixing	Sample random 5s window from unlabeled Pantanal soundscape. Mix at SNR ~ Uniform(−5, +10) dB. Backgrounds are used for training backbone.
[6] Waveform mixup	With prob 0.5: sample second clip. Mix $x = \lambda \cdot x_1 + (1-\lambda) \cdot x_2$, $\lambda \sim \text{Beta}(0.4, 0.4)$. Blend labels: $y = \lambda \cdot y_1 + (1-\lambda) \cdot y_2$ (species or background).
[7] Mel spectrogram	$n_mels=128$, $hop_length=320$ (10ms), $n_fft=1024$, $fmin=50$, $fmax=14000$. Result: 128×500 per 5s window. Log-scale.
[8] SpecAugment (P3)	After P1–P2 are validated: freq masking 1–2 bands ($\leq 10\%$ mel bins), time masking 1 block ($\leq 8\%$ frames). Conserve spectral structure.

3.3 Feature Extraction Recommendations

- **Sample rate:** 32 kHz mono — already the standard in the training data.
- **Window size:** 5 seconds (fixed) — matches the prediction unit exactly.
- **Mel bins:** 128 — sufficient for bird and frog frequency resolution; increase to 256 only if non-bird classes are the bottleneck.
- **fmin:** 50 Hz — captures low-frequency frog calls ($\text{Amphibia} \geq 200 \text{ Hz}$) and mammal calls without excessive low-frequency noise.
- **fmax:** 14,000 Hz — captures most insect sonotypes (peak 4–12 kHz) without wasting bins above 14 kHz where little target signal exists.
- **Pre-compute spectrograms** before training where possible. On-the-fly extraction from 35K OGG files causes significant I/O bottleneck.

3.4 Label Construction

- **Clip clips → window labels:** assign `primary_label` + `secondary_labels` to all non-silent windows. Accept label noise; energy gating handles the worst cases.
- **Soundscape windows (labeled):** use exact per-window expert labels. These are the gold standard — prioritize them in the loss or oversample them.
- **Unlabeled soundscape windows (background mix source):** label as all-zeros when used as background. Do NOT infer labels from clip-level overlap.
- **Rare-species oversampling:** weight each labeled soundscape window by $1/\text{sqrt}(\text{rarest_class_freq})$ to ensure all 234 classes appear in every epoch.
- **Soft labels for mixup:** BCE loss must support continuous $[0,1]$ targets. Confirm this in your chosen framework before training.

§4 — Validation Strategy

A realistic validation scheme is the single most important infrastructure decision. An optimistic CV scheme means every experiment looks better than it is, and real leaderboard position is a surprise. BirdCLEF competitions are notorious for site-level acoustic leakage inflating OOF scores.

4.1 Why Random K-Fold Is Wrong Here

- **Author leakage:** Top 5 authors contribute 40%+ of clips. A random split puts clips from the same recording session in both train and validation. The model partially memorizes recorder fingerprints.
- **Location leakage:** train clips have GPS coordinates. Nearby recordings sound similar (same forest, same background noise). A random split leaks location-level acoustic context.
- **Site leakage in soundscapes:** site S22 dominates labeled windows (64%). A random split of soundscape windows will have both train and validation from S22, making the model appear to generalize when it is just recognizing site acoustics.
- **Temporal leakage:** same recorder makes multiple trips to same site in same year. Near-duplicate recording clusters (EDA Section G3) will leak across random splits.

4.2 Recommended Validation Scheme

Scheme	How to split	When to use
Site-stratified 5-fold (clips)	Group clips by GPS cluster (K-means on lat/lon → 20 clusters).	Each fold holds out a different geographic cluster. Primary CV for clip-based programs.
Author-stratified 5-fold (clips)	Group clips by author/recorder. Each fold holds out complete author groups.	Cross-check: should give similar result to site-stratified.
Soundscape site holdout	Reserve 3 soundscape sites (not S22) as held-out evaluation. Train on all other clips.	Primary CV for S22-based programs. The held-out sites are not concentrated in any one year.
Taxonomic-stratified check	Ensure each fold contains examples from all 5 taxon classes.	Check that temporal and site leakage are not concentrated in any one taxon.

Key rule: Report OOF AUC separately per taxonomic class (Aves, Amphibia, Insecta, Mammalia, Reptilia). A high overall macro AUC driven by bird classes while frog/insect AUC is near 0.5 is a false positive. The leaderboard will expose this.

§5 — Augmentation Plan (Evidence-Based)

Each augmentation below is tied to a specific EDA finding. Generic augmentations not supported by evidence are excluded. The list is ordered by implementation priority.

Priority	Augmentation	Problem it addresses (EDA source)	Caveat
P1 — Must have	Energy gating (preprocessing)	14% mean silence; some clips 90% silent. Clip-level labels assigned to silent windows → false positives. (Section E3, F3)	Threshold tuning: test −45, −40, −50 dBFS
P1 — Must have	Peak normalization (preprocessing)	600× loudness variance across recordings. Without normalization, model learns level as a species cue. (Section E3)	Apply per-clip, before any windowing.
P1 — Must have	Background mixing (soundscape overlay)	Clip→soundscape domain gap is the primary train/test mismatch. Real Pantanal noise backgrounds are available from 10,592 unlabeled soundscapes. (Section G5, F1)	Pre-screen soundscapes for extreme artifacts.
P1 — Must have	Waveform mixup ($\lambda \sim \text{Beta}(0.4, 0.4)$)	Training clips are mostly single-label. Expert soundscape windows show 87% have 2+ co-active species. (Section F2, C)	Requires soft-label BCE loss. Confirm frame-level labels.
P2 — High value	Rare-species oversampling	28 zero-clip species; many with < 5 clips. Standard sampling ignores them for entire epochs. (Section B, C)	Weight = $1/\sqrt{\text{class_freq}}$. Watch common class bias.
P3 — Medium	SpecAugment (conservative)	Top-5 author recorder fingerprints = 40%+ of data. Frequency masking breaks EQ-pattern memorization. (Section G2)	Max 10% freq bins, max 8% time frames. Test on real data.
P3 — Medium	Soft saturation (tanh clipping sim)	~8% of clips have clipping artifacts. Model should be robust to saturation distortion. (Section E3)	Apply to a random fraction (20–30%) of clips.
P3 — Low	Time-of-day feature (not an augmentation)	Dawn chorus peak in soundscapes; species activity is time-conditional. Test row_id encodes datetime. (Section E4)	Conditionals: $\sin(2\pi \cdot \text{hour}/24)$, $\cos(2\pi \cdot \text{hour}/24)$

5.1 Augmentations Explicitly Excluded

Excluded augmentation	Reason
Large pitch shift ($\pm 4+$ semitones)	Frog and insect calls are pitch-specific for species ID. Large shifts change species identity.
Time stretching > $\pm 20\%$	Call temporal rhythm (rate, duty cycle) is species-diagnostic, especially for insects.
Additive Gaussian / white noise	Pantanal background is structured (water, wind, co-species). Use real soundscape backgrounds.
Spectrogram CutMix	Pasting spectrogram patches creates acoustically impossible frequency chimeras. Waveform mixup is preferred.
Random crop to < 2.5s	Creates training examples shorter than half a prediction window. Label assignment becomes meaningless.
Frequency shifting (full spectrum)	Harmonic structures are key species discriminators. Shifting destroys harmonic ratios.

§6 — Prior Data Decision

This section answers: which prior BirdCLEF competition data should be used, in what capacity, and what should be avoided. The answer is not binary — different datasets serve different purposes.

Dataset	Geography	Taxon overlap	Domain match	Recommended use	Risk
BC2025 clips + soundscapes	Pantanal, Brazil	High (same region)	Excellent — same species	Use directly for training after deduplication	Possible overlap with BC2026 soundscapes
BC2024 clips	Multiple regions	Moderate	Poor — different regions	Backbone pre-training only	Hurts calibration if mixed directly
BC2023 / BC2022 clips	Global / India / Africa	Low	Very poor	Backbone pre-training only if needed	Geographic and species mismatch
BC2021 clips	Cornell / North America	Very low	Very poor	Skip or use only for very broad pre-training	Highly mismatched; may hurt
BC2024 soundscapes (non-Pantanal)	Other regions	Low	Very poor	Do NOT use as domain training data	Confuses domain adaptation; hurts soundscapes

Decision: Use BC2025 as in-domain training data (high priority). Use BC2021–2024 clips for backbone pre-training only. Do NOT use non-Pantanal soundscapes as soundscape training data.

Mandatory step before combining any prior data: deduplicate by Xeno-Canto URL. Prior competitions draw from the same XC pool. Duplicate clips in both train and val will inflate OOF AUC.

§7 — Risk Register

Risk	Severity	Source (EDA)	Mitigation
Clip→soundscape domain gap	CRITICAL	G5, F1	Background mixing (H1). Site-stratified CV (H4). Monitor per-site AUC during training
Label noise from clip windowing	HIGH	E3, F3	Energy gating (H3). Discard silent windows. Accept some noise for non-silent windows
28 zero-clip species (25 insect sounds + 3 frog sounds)	HIGH	B5, C	Oversample labeled soundscape windows (H3). Accept near-zero AUC on insect sounds
Site S22 dominance in labeled soundscapes (64% of windows)	LOW	C4	Hold out non-S22 sites for evaluation. Do not train final model exclusively on S22-dominant sites
Author/recorder fingerprint memorization	MEDIUM	G2	Author-stratified CV (4.2). SpecAugment frequency masking (P3). Monitor OOF gap
Seasonal mismatch (test = Feb 2025 → start of dry season)	MEDIUM	D5, E4	Add time-of-day and month features (H10). Weight training on wet-season data slightly
Rating-zero clips degrading training (30% of clips are 0)	LOW	A5, E4	Audit: listen to 50 random rating-0 clips. If systematic noise → train with quality-stratified
XC duplicate clips across BC2025-MBO2026 inflating OOF	MEDIUM	A5	Deduplicate by XC URL before combining datasets.
CPU inference timeout on complex models	HIGH	A §12	Design around ≤ EfficientNet-B2 or ConvNext-Tiny equivalent. Profile inference time
Macro AUC masking poor non-bird taxon performance	MEDIUM	A §8	Track OOF AUC per taxon. Target: all taxon classes ≥ 0.70. Non-bird AUC below 0.50

§8 — Phased Experiment Plan

Experiments are sequenced to maximize information gain per GPU-hour. Each phase must produce a validated OOF AUC before moving to the next. Do not stack unvalidated interventions.

Phase	Dates (approx)	Goal	Exit criterion
Phase 0: Infrastructure	Weeks 1–2 (by Apr 26)	• Clip windowing + energy gating pipeline • Site-stratified 5-fold splits • Mel spectrogram extraction (batched) • Baseline EfficientNet-B0 flat 234-class • OOF AUC tracking per taxon	Reproducible baseline OOF AUC > 0.6 Inference time < 60 min on CPU
Phase 1: Core Augmentation	Weeks 2–3 (by May 3)	• Peak normalization (H7) • Background mixing P1 (H1) • Waveform mixup P1 (H2) • Rare-species oversampling (H3)	+0.03 OOF AUC vs Phase 0 Non-bird taxon AUC > 0.60
Phase 2: Data Expansion	Weeks 3–4 (by May 10)	• Add BC2025 data + dedup (H5) • Add Neotropical XC for zero-clip species • SpecAugment P3 validation	+0.02 OOF AUC vs Phase 1
Phase 3: Architecture	Weeks 4–5 (by May 17)	• Multi-taxon head experiment (H8) • Backbone pre-training on BC2021–2024 (H6) • Time-of-day conditioning (H10)	+0.02 OOF AUC vs Phase 2 Decide: multi-taxon head yes/no
Phase 4: Semi-supervised	Weeks 5–6 (by May 24)	• Pseudo-labeling on unlabeled soundscapes (H9) • Two-stage insect sonotype detection (H11) • Model ensemble (2–3 seeds minimum)	+0.01–0.02 OOF AUC Insect sonotype AUC > 0.55
Phase 5: Final polish	Weeks 6–8 (by Jun 3)	• Inference optimization (ONNX / torch.compile) • Prevalence post-processing experiment (H12) • Final ensemble submission • Submission slot management	Submission time < 85 min Final ensemble > Phase 4 OOF

Checkpoint rule: If a phase does not meet its exit criterion after 2 attempts, diagnose root cause before moving forward. Do not advance to the next phase on faith — every phase builds on the previous one.

§9 — Open Questions & Gaps

These are things that cannot be resolved from EDA alone and require either direct experimentation or access to the hidden test set.

Question	Why it matters	How to resolve
Which species appear in the test set?	Class 23 is present in test but skipped from macro coverage. Treating all 23 classes as a posteriori	Check known test files covering ground truth
How many test soundscapes are there?	Determines inference time budget per soundscape	Check sample_submission.csv count. Estimate: ~50–200 soundscapes
What quality are the rating-zero clips?	36% of training clips are unrated. If many are low quality, 50% should be dropped	Visualize and inspect unrated clips
Are BC2025 soundscapes from the same physical sites as BC2026?	BC2026 includes BC2025 and BC2026 test data possible. If site IDs overlap: exclude	Check BC2026 test data
Are insect sonotypes actually distinguishable?	Some insect sonotypes are acoustically ambiguous	Spectrogram visualization of 25 classes
What is the test soundscape recording frequency?	Fig 1.2025 is 10 kHz	Interpret and compare with seasonal competition events
Does BC2026 evaluation skip truly zero-clip classes?	The positive class determines the 234?N for macro coverage	Check if test set has competition AQA. As of the 2023 classes

Report Summary

This report synthesizes EDA Sections A–I into a concrete, prioritized strategy. The three most important takeaways:

- **The domain gap (clip → soundscape) is the #1 problem.** Background mixing with unlabeled Pantanal soundscapes (H1) is the highest-ROI single intervention. Do this before anything else.
- **Validation scheme determines everything.** Site-stratified 5-fold CV is mandatory. Random splits will produce inflated OOF that doesn't match LB.
- **Non-bird taxa are the leaderboard differentiator.** Frogs, insects, and mammals are 30% of targets. Track their AUC separately. Most competitors will under-optimize for them.